

Global Physical Prior Based Fluid Reconstruction for VR/AR

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Figure 1: A cup of hot tea: Smoke beyond the cup in the real world (top) is captured in a video, the density fields and velocity fields are reconstructed by our algorithm, and finally re-rendered in a AR application (bottom).

ABSTRACT

Fluid is a common natural phenomenon and often appears in various VR/AR applications. Several works use sparse view images and integrate physical priors to improve reconstruction results. However, existing works only consider physical priors between adjacent frames. In our work, we propose a differentiable fluid simulator combined with a differentiable renderer for fluid reconstruction, which can make full use of global physical priors among long series. Furthermore, we introduce divergence-free Laplacian eigenfunctions as velocity bases to improve efficiency and save memory. We demonstrate our method on both synthetic and real data and show that it can produce better results.

Index Terms: Computing methodologies—Modeling and simulation—Model development and analysis—Modeling methodologies

1 INTRODUCTION

In many VR/AR applications, like fire drills, fluid is an essential component, and the effect of simulation drills benefits a lot from realistic fluid behaviors. However, fluids in nature have unpredictable, complex motions that are difficult to reproduce by numerical simulation. Fluid capturing, in another way, tries to retrieve fluid properties (e.g., advected density and velocity) directly from real-world observation, which is a reconstruction problem and has been actively studied in computer graphics and computer vision. The reconstructed flows

can be further used for artistic purposes such as guiding, control, stylization, and VR/AR applications.

Multi-view videos are commonly used for reconstructing fluid flows, which require several cameras to shoot simultaneously at different fixed views. However, due to complex setup and difficult calibration, the number of views is often sparse. This makes the fluid reconstruction problem highly under-determined and makes it hard to obtain natural and realistic flows. To tackle this problem, many recent works integrate physical priors (i.e., Navier-Stokes equations) and density advection into the reconstruction process and achieve better results. Nevertheless, most of existing methods only involve temporally local physical priors, probably considering the simplicity or efficiency of solving, which may lead to a suboptimal solution that derives from the physical model. Given this, we propose a fully differentiable framework for fluid reconstruction to make full use of physical priors. Our framework consists of a differentiable fluid simulator (DFS) to connect 3D fluid fields of the entire sequence and a differentiable renderer (DR) to connect images with 3D density fields of fluid. Based on the adjoint method, all necessary derivatives in our framework can be computed in only one backward pass, and this makes it easy and efficient to solve using gradient-based optimization algorithms.

2 OUR APPROACH

2.1 DFS with Divergence-free Laplacian Eigenfunctions

Our DFS involves Eulerian grids. The forward calculation process of our DFS consists of the following steps: advect velocity, project velocity, add divergence-free force, advect density and add density inflow. The backward derivatives are calculated in reverse order. Here we only focus on the velocity projection step because other steps are relatively trivial, and derivatives can be properly derived by automatic differentiation tools. However, traditional velocity projection, i.e., pressure projection, involves solving a linear equation iteratively. When we calculate the derivatives using the adjoint method [5], the accuracy of derivatives is affected by the accuracy of

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the linear solver in this step. So one needs to trade time for accuracy. In our work, we introduce divergence-free Laplacian eigenfunctions [1] as bases of 3D divergence-free velocity fields, and pressure projection can be replaced by directly projecting the velocity fields on these bases to achieve high accuracy of derivatives.

Laplacian eigenfunctions form a complete orthogonal basis tailored to a specific domain with an appropriate boundary condition, which is derived from the eigenfunctions of the vector Laplacian operator. In our work, we use a rectangular domain to reconstruct fluids and use no-slip Neumann boundary conditions. In this case, the analytic form of our bases consists of sine and cosine functions, so we can speed up projection using a mix of discrete sine/cosine transform. Furthermore, representing divergence-free velocity fields with basis coefficients requires only two-thirds of the memory usage.

2.2 DR with Soft Visual Hull Constraint

In our work, to speed up the rendering process, we adopt single scattering with ambient light to approximate multiple scattering. We approximate all integrals by raymarching. To reduce the time complexity, we precompute the outgoing radiance at each voxel center of a given density field and then perform tri-linear interpolation at each sample point. The derivative backpropagation process of our differentiable rendering is derived and implemented manually. For the stability of derivatives, we do not consider derivatives in precomputed outgoing radiance.

Visual hull constructs a conservative estimation where may exist a non-zero density value built from input multi-view images. It is often used as a hard constraint to get rid of density residuals outside. However, in our reconstruction framework, density fields are not allowed to be optimized directly. Thus we consider it a soft constraint. To further avoid extra regularization terms, we propose a novel but simple derivative-related soft visual hull application strategy. For each input view, we limit the derivatives outside the visual hull to non-negative so that density values do not increase further. Note that there must exist at least one input view where those density values are decreased because they are out of the visual hull.

2.3 Global Physical Priors Based Reconstruction

The final goal of our framework is to reconstruct density fields and velocity fields of all frames that match the input sparse-view videos and adhere to physical priors imposed on the entire sequence. But to make better use of physical priors, we do not optimize density and velocity fields directly. Otherwise, we optimize quantities that will change the behaviors of fluid, i.e., density inflow fields and force fields. We optimize the problem using a gradient descent algorithm. We find that density fields and density inflow fields can share the same memory space by checkpointing to further trade time for space. So are velocity fields and force fields. Spectral-domain multi-scale strategy is applied to force fields to reconstruct large-scale motions first. At the same time, derivatives should decay exponentially with frames to avoid gradient explosion over long sequences.

3 RESULTS AND DISCUSSIONS

To examine the effectiveness of our method, we use smoke as our reconstruction target media. We first verify the effectiveness of our method by conducting qualitative and quantitative experiments on synthetic data. Specifically, we design two ablation experiments to verify global physical priors and divergence-free Laplacian projection, respectively, and compare our results with state-of-the-art methods: ScalarFlow (SF) [2] and Global Transport (Glob-Trans) [3]. The experiment results are shown in Figure 2 and Table 1. The AR display of our reconstructed results on the real data is shown in Figure 1.

This paper presents a method to recover the density and velocity of a fluid from an input video sequence. Our reconstruction process

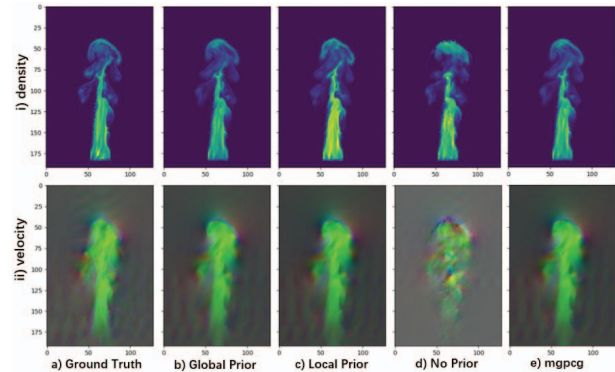


Figure 2: The slices of density fields (top) and velocity fields (bottom). a) The ground truth is synthetic data simulated by fluid solver, b,c,d) Reconstruction results of our method with global physical priors, local physical priors and no physical priors respectively. e) Reconstruction results with global physical priors but replacing the Laplacian projection with traditional pressure projection.

Metric	denNORM	denSSIM	velNORM	velSSIM
Laplace	1.27755	0.61634	0.18893	0.26844
mgpcg [4]	1.28013	0.61894	0.18698	0.27469
ScalarFlow [2]	2.90896	0.27691	0.18645	0.45389
Glob-Trans [3]	14.95633	-0.32251	0.96710	-0.54347

Table 1: Quantitative comparison results of density and velocity fields. The first two lines are the quantitative comparison for the ablation experiments, and the last two lines are the quantitative comparison with ScalarFlow [2] and Glob-Trans [3].

satisfies global physical priors rather than just considering derivatives between adjacent frames. At the same time, divergence-free Laplacian eigenfunctions are introduced to project velocity instead of using traditional pressure projection to improve efficiency and save memory. Furthermore, we trim the derivatives in the reconstruction process by several strategies to make reconstruction results better. Experimental results show that our method can improve the accuracy of the fluid reconstruction and make the obtained velocity field more natural. At the same time, it is possible to further improve our method. First of all, we can not reconstruct the fluid with rigid bodies. Second, a more sophisticated renderer can be introduced. Third, quasi-Newton optimization can be used to speed up convergence.

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